

COMICCRAFT – AI POWERED COMIC STORY GENERATOR

Phase 1: Brainstorming & Ideation

1. Project Title

ComicCraft – AI Powered Comic Story Generator

2. Problem Statement

Creating comics traditionally requires multiple skills such as story writing, character design, illustration, scene creation, dialogue writing, and page layout design. This process can be time-consuming, expensive, and inaccessible to people without artistic expertise.

Many aspiring storytellers have creative ideas but lack the technical or artistic skills required to transform those ideas into visually appealing comics.

ComicCraft addresses this challenge by leveraging Generative AI technologies to automatically generate comic stories, dialogues, illustrations, and downloadable comic books from simple user prompts.

3. Background

With the rapid advancement of Generative AI, Large Language Models (LLMs) and AI image generation systems can now create high-quality textual and visual content from natural language prompts.

ComicCraft combines text generation and image generation technologies to automate comic creation. Users only need to provide a story idea, character information, preferred tone, and art style. The system then generates a complete comic consisting of multiple panels, narration, dialogue, illustrations, and PDF export functionality.

This project demonstrates the practical application of Generative AI in creative content generation and digital storytelling.

4. Project Overview

ComicCraft is an AI-powered web application that transforms user-provided story prompts into complete comic books.

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The application performs the following tasks:

- Generates a structured comic storyline
- Creates multiple comic panels
- Generates dialogues and narration
- Produces AI-generated illustrations
- Combines story and images into a comic layout
- Exports the final comic as a downloadable PDF

The system aims to make comic creation accessible to everyone regardless of artistic or technical background.

5. Objectives

The primary objectives of ComicCraft are:

1. To automate comic creation using Generative AI.
 2. To generate engaging comic stories from simple prompts.
 3. To create visually appealing illustrations for each comic panel.
 4. To provide a user-friendly web interface.
 5. To generate downloadable PDF comics.
 6. To reduce the effort required for comic development.
 7. To demonstrate the integration of multiple AI technologies within a single application.
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6. Target Users

The target users of ComicCraft include:

- Students
- Story Writers
- Content Creators
- Comic Enthusiasts
- Educators
- Digital Artists

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- Marketing Teams
 - Social Media Creators
 - Beginners interested in AI-powered content creation
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7. Existing System

Current comic creation methods generally involve:

- Manual story writing
- Manual character design
- Manual illustration
- Graphic design software
- Professional comic development tools

Limitations of existing systems:

- Time-consuming
 - Expensive
 - Requires artistic expertise
 - Difficult for beginners
 - High production effort
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8. Proposed System

ComicCraft introduces an AI-powered automated comic generation system.

The proposed system allows users to:

1. Enter a story prompt.
2. Specify character details.
3. Select tone and art style.
4. Generate a comic storyline.
5. Generate comic panel images.
6. Preview the comic.
7. Download the comic as a PDF.

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Benefits of the proposed system:

- Faster comic creation
 - Reduced manual effort
 - User-friendly interface
 - AI-generated illustrations
 - Automated storytelling
 - Accessible to non-artists
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9. Key Features

Story Generation

Uses Large Language Models to generate a structured comic storyline.

Dialogue Generation

Creates character conversations and narration automatically.

Comic Panel Creation

Generates multiple panels for a complete comic story.

AI Illustration Generation

Creates visual content using image-generation models.

Comic Layout Builder

Combines text and images into comic format.

PDF Export

Allows users to download the generated comic.

Responsive Web Interface

Provides a simple and interactive user experience.

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10. Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript
- Jinja2 Templates

Backend

- Python
- FastAPI
- Pydantic

AI Technologies

- Google Gemini Flash
- Google Gemini Pro

Image Generation

- FLUX
- Stable Diffusion
- Hugging Face Inference API

Export Tools

- FPDF
- Pillow

11. Proposed Workflow

Step 1: User enters a story prompt.

Step 2: System validates the input.

Step 3: Gemini Flash generates a 5-panel comic outline.

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Step 4: Gemini Pro expands the outline into a detailed comic story.

Step 5: AI image generation creates illustrations for each panel.

Step 6: Comic layout builder combines text and images.

Step 7: PDF generator creates the final comic book.

Step 8: User previews and downloads the comic.

12. System Architecture

User Input



Frontend Interface



FastAPI Backend



Gemini Flash (Outline Generation)



Gemini Pro (Story Generation)



Image Generation Model



Comic Layout Builder



PDF Generator



Comic Preview



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13. Scope of the Project

The project scope includes:

- Story generation
- Comic panel generation
- Dialogue creation
- Image generation
- PDF export
- Web-based user interface

The project does not currently include:

- Video generation
- Animation generation
- Voice synthesis
- Real-time collaboration

14. Expected Outcomes

Upon completion, ComicCraft will:

- Generate complete comic stories automatically.
- Produce AI-generated illustrations.
- Create downloadable PDF comics.
- Reduce comic creation time significantly.
- Provide an engaging user experience.
- Demonstrate real-world Generative AI integration.

15. Future Enhancements

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1. Multi-character story generation
 2. Manga-style comics
 3. Animated comic generation
 4. Voice narration
 5. Mobile application
 6. Character memory system
 7. Story continuation mode
 8. Multi-language support
 9. Cloud deployment
 10. AI-assisted comic editing
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16. Conclusion

ComicCraft is an innovative Generative AI application designed to simplify comic creation through automated storytelling and image generation. By combining Large Language Models and AI image-generation technologies, the system enables users to transform simple ideas into complete comic books with minimal effort. The project demonstrates the practical use of modern AI technologies in creative content generation and has the potential to significantly enhance digital storytelling experiences.